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| LLM2025                                                             | DIR-LLM      | CC1      |
| Coordinated by:                                                     | Tom QUIBLIER | 2 months |
| Course : Theory and Practical Applications of Large Language Models |              |          |

# DIR-LLM: Detection of Illusory Reasoning in Large Language Models

## I Pre-proposal context, positioning and objective(s)

### a Context

Large Language Models (LLMs) have recently achieved remarkable performance in natural language generation, problem solving, and apparent reasoning. Thanks to large-scale training and Transformer-based architectures, these models are now widely used in applications such as education, decision support, scientific assistance, and programming. Their ability to generate detailed, structured explanations gives the impression of strong reasoning capabilities.

However, despite these impressive results, LLMs suffer from well-known limitations, in particular the phenomenon of hallucination. Factual hallucinations, which correspond to incorrect or fabricated statements about the world, have been extensively studied. A more subtle and potentially more dangerous phenomenon is now gaining attention: reasoning illusions.

Reasoning illusions refer to situations in which a model produces an apparently coherent, well-structured, and convincing chain of reasoning that is nevertheless logically invalid. The final conclusion may be false or unjustified even though the explanation appears sound on the surface. Such errors are particularly problematic because they are difficult for non-expert users to detect and can induce unjustified trust in the model's reasoning abilities.

In sensitive domains such as education, science, law, or decision-making systems, these reasoning illusions may lead to serious consequences. It is therefore essential to better understand this phenomenon and to develop automatic tools capable of identifying it.

### a1 Positioning to state of the art

Factual hallucinations in LLMs have been widely investigated in the literature. Numerous approaches have been proposed to mitigate them, including retrieval-augmented generation, automatic fact-checking, confidence calibration, and reinforcement learning from human feedback.

Regarding reasoning, many studies have evaluated the performance of LLMs on logical, mathematical, and symbolic tasks. Chain-of-Thought prompting has notably improved the ability of models to decompose complex problems into intermediate steps and to produce more explicit reasoning. Nevertheless, recent works have shown that these explanations often rely on superficial statistical patterns rather than on genuine logical understanding.

Some recent studies have begun to investigate the failures of LLM reasoning by evaluating their sensitivity to logical fallacies, inconsistencies, and prompt variations. Other works explore the idea of using a second LLM as a judge or critic of the generated reasoning. However, there is currently no standardized methodological framework to formally define, detect, and systematically evaluate reasoning illusions.

As a result, while factual hallucinations are now relatively well understood, the automatic detection of reasoning illusions remains a largely open research problem, which motivates the originality and relevance of this project.

## b Project's objectives and Methodology

### b1 Objective 1: Formal definition and typology of reasoning illusions

The first objective of the project is to provide an operational definition of reasoning illusions in the context of LLMs. Based on existing literature in logic, artificial intelligence, and reasoning analysis, we will propose

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a simple but meaningful typology of such illusions. This typology may include, for instance, inference errors, internal contradictions, false causal relationships, over-generalizations, and post-hoc fabricated justifications. This objective is mainly theoretical but will serve as the conceptual foundation for the experimental work.

## b2 Objective 2: Construction of a controlled evaluation dataset

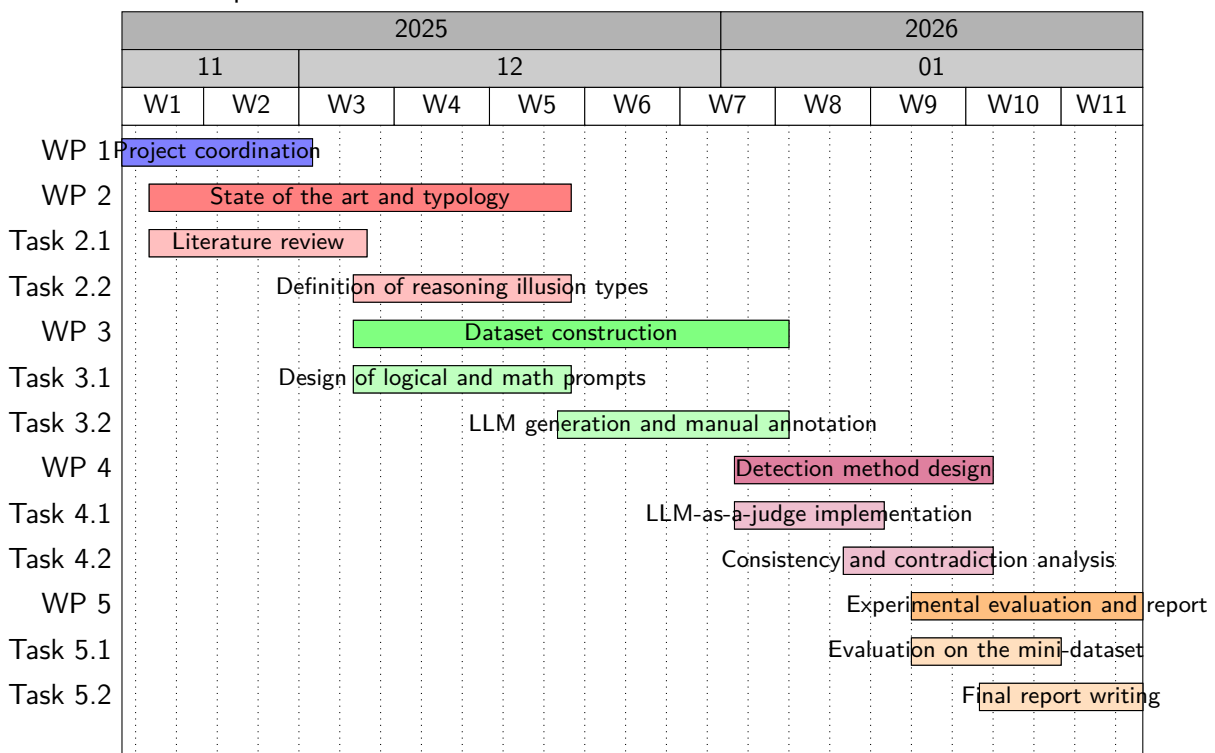
The second objective is to build a small evaluation dataset designed to trigger different types of reasoning illusions. This dataset will include simple mathematical problems, formal logical reasoning tasks (such as syllogisms, implications, and negations), as well as open-ended questions likely to induce plausible but incorrect justifications. The model outputs will be manually annotated according to the proposed typology. This dataset will constitute the experimental basis for evaluating the detection methods.

## b3 Objective 3: Design of an automatic detection method

The third objective is to propose one or several automatic methods for detecting reasoning illusions. Several complementary strategies will be explored. A first approach will rely on using a second LLM as a logical evaluator or critic of the generated reasoning. Another approach will be based on comparing multiple chains of reasoning produced for the same input in order to identify inconsistencies. Finally, an internal analysis of the generated explanations will be conducted to automatically detect contradictions or logical breaks. These different approaches will be empirically compared.

## b4 Objective 4: Experimental evaluation and analysis of limitations

The final objective is to empirically evaluate the proposed detection methods on the constructed mini-dataset. Detection rates, false positives, and false negatives will be analyzed. Special attention will be paid to ambiguous cases where the boundary between valid reasoning and illusion is unclear. This phase will also allow us to discuss the methodological and computational limitations of the proposed system and to identify directions for future improvements.



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